SUPPLEMENTARY INFORMATION

ReRAPTOR: Reverse-Objective RAPTOR for Time-Budgeted Value-Maximizing Transit Itinerary Planning

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These tables provide a visible numerical source for detailed values cited in the manuscript. Unless noted otherwise, a gap is ReRAPTOR minus NSGA-II, and a 95% confidence interval (CI) is the crossed block-bootstrap interval defined in Section 5.1. Values are rounded only for display; calculations use the unrounded source files.

In the tables, W/T/L means wins/ties/losses relative to the stated comparator. `CapHit` is the percentage of tasks stopped by the 60-s timeout or, for the no-pruning configuration, the 1,000,000-label cap. "Generation" is the solver wall-time boundary defined in Section 5.1 of the manuscript.

Supplementary Table S1. Overall paired interpretation and reliability

Estimand or diagnostic	Tasks	Estimate	95% CI
Stop Score gap, all-task failure-zero policy	600	69.899	[8.054, 130.917]
Reward-equivalent gap, 69.899/50.5	600	1.384	[0.159, 2.592]
Unique scored-stop-count gap	600	1.320	[0.535, 2.092]
Selected reward per scored stop gap	581	−1.914	[-3.247, -0.654]
Score gap, all five NSGA-II runs feasible	452	4.708	[-63.024, 67.875]
Score gap, mean over feasible NSGA-II runs only	581	−3.059	[-71.301, 62.440]

Reliability diagnostic	Count or value
ReRAPTOR wins / ties / losses across matched tasks	380 / 49 / 171
ReRAPTOR positive feasible routes	600 / 600
Base queries with both higher ReRAPTOR score and lower time	92 / 120
NSGA-II algorithm-seed failures	288 / 3000
Tasks affected by at least one NSGA-II failure	148 / 600
Tasks on which all five NSGA-II runs failed	19 / 600
Conditional mean loss when ReRAPTOR loses	236.423
Mean gap in the worst task decile	−556.350

Source: `paper/results/exp1_algorithm_comparison/score_interpretability_summary_20260723.csv`

Source: `paper/results/exp1_algorithm_comparison/underperformance_summary_20260723.csv`

Supplementary Table S2. Budget and reward-seed robustness

Panel A: Budget blocks

Budget	Tasks	ReRAPTOR	NSGA-II	Gap	95% CI	W/T/L
2h	150	398.613	327.980	70.633	[34.860, 118.036]	79/46/25
4h	150	803.747	656.504	147.243	[93.412, 204.994]	116/1/33
6h	150	988.907	965.940	22.967	[-75.432, 117.743]	87/0/63
8h	150	1298.487	1259.733	38.753	[-105.307, 167.186]	98/2/50

Panel B: Generation time and feasible-route reliability by budget

Budget	ReRAPTOR (ms)	NSGA-II (ms)	NSGA/ReRAPTOR	ReRAPTOR positive feasible
2h	82.656	23,221.838	280.95	150/150
4h	2,042.989	33,730.696	16.51	150/150
6h	3,166.403	40,131.436	12.67	150/150
8h	15,392.676	45,455.155	2.95	150/150

Panel C: Reward-seed blocks

Reward seed	Tasks	Gap	95% CI	W/T/L
20260323	120	61.660	[-17.146, 136.594]	80/6/34
20260324	120	117.615	[48.582, 192.246]	74/8/38
20260325	120	80.475	[22.521, 134.540]	70/13/37
20260326	120	41.545	[-29.599, 107.711]	83/11/26
20260327	120	48.200	[-35.496, 128.903]	73/11/36

Source: `paper/results/exp1_algorithm_comparison/underperformance_summary_20260723.csv`

Source: `paper/results/exp1_algorithm_comparison/exp1_score_guided_multiseed_summary_canonical_20260727.md`

Supplementary Table S3. Origin, departure-time, and long-budget subgroups

Panel A: Origin and departure-time blocks

Grouping	Level	Tasks	Mean gap	95% CI	W/T/L
Origin	Fukudaimae	60	252.967	[117.360, 431.813]	46/10/4
Origin	Hakata	60	29.223	[-79.917, 137.613]	30/3/27
Origin	Kaizuka	60	147.130	[57.720, 267.167]	42/6/12
Origin	Kashii	60	153.560	[15.450, 317.707]	47/4/9
Origin	Meinohama	60	192.007	[127.910, 255.740]	49/7/4
Origin	Nishijin	60	112.477	[-7.590, 255.830]	43/3/14
Origin	Noke	60	−225.693	[-361.123, -88.403]	25/2/33
Origin	Ohori-koen	60	−17.383	[-161.437, 116.793]	29/5/26
Origin	Tenjin	60	72.363	[-17.330, 153.910]	38/4/18
Origin	Yakuin	60	−17.660	[-227.190, 131.063]	31/5/24
Departure	07:00	200	32.125	[-83.329, 149.383]	114/13/73
Departure	10:00	200	120.373	[14.141, 207.916]	144/15/41
Departure	14:00	200	57.199	[-30.363, 130.670]	122/21/57

Panel B: Origin–budget cells highlighted in the manuscript

Origin	Budget	Tasks	Mean gap	95% CI
Noke	6h	15	−468.920	[-683.387, -211.960]
Noke	8h	15	−687.200	[-987.920, -385.373]
Ohori-koen	6h	15	−160.400	[-332.600, 36.795]
Ohori-koen	8h	15	−186.293	[-470.987, 68.720]
Yakuin	6h	15	−149.867	[-471.587, 78.213]
Yakuin	8h	15	−127.720	[-579.387, 153.813]

Source: `paper/results/exp1_algorithm_comparison/underperformance_summary_20260723.csv`

Source: `paper/results/exp1_algorithm_comparison/underperformance_origin_budget_20260723.csv`

Supplementary Table S4. Budget and taskwise ablation checks

Panel A: Full versus NoBeam by budget

Config.	Budget	Score	Generation (ms)	Labels	CapHit	Timeout
Full	2h	398.613	163.615	2,244	0.00%	0.00%
NoBeam	2h	411.793	1,424.927	18,436	0.00%	0.00%
Full	4h	803.747	4,364.737	47,322	0.00%	0.00%
NoBeam	4h	799.633	48,799.429	594,929	70.67%	70.67%
Full	6h	988.907	6,852.295	63,081	0.00%	0.00%
NoBeam	6h	815.187	60,984.728	766,424	100.00%	100.00%
Full	8h	1298.487	29,771.468	253,142	18.67%	18.67%
NoBeam	8h	796.253	60,957.951	778,063	100.00%	100.00%

At 2h, the directly calculable NoBeam/Full time ratio is 1424.927/163.615 = 8.71; NoBeam is 13.180 points higher. Its 6h and 8h timeout rates are both 100%.

Panel B: Full versus NoNeg across matched tasks

Comparison	Tasks	Mean score change	W/T/L	Interpretation
NoNeg minus Full	600	+0.033	6/594/0	no generic benefit established

Source: `paper/results/exp2_ablation_study/exp2_canonical_by_budget_20260725.csv`

Source: `paper/results/exp2_ablation_study/exp2_canonical_paired_vs_full_20260725.csv`

Supplementary Table S5. Balance and hyperparameter sensitivity

Panel A: Experiment 1 balance diagnostics

Diagnostic	ReRAPTOR	NSGA-II
Mean ideal-point distance across 120 base queries	0.131534	0.741286
Queries with higher ReRAPTOR score and lower time	92 / 120	

Panel B: One-at-a-time sensitivity (600 tasks per configuration)

In the default row, coefficients 10000/5/0.1/1 are, respectively, the reward, SPH, effective-time, and Negative-penalty-scale coefficients.

Factor	Value	Mean score	Paired change [95% CI]	W/T/L	Mean labels	CapHit
Default	W50; cap 30/20; coefficients 10000/5/0.1/1	872.438	0.000 [0, 0]	0/600/0	93,323	3.50%
Beam width	10	872.125	−0.313 [−0.827, 0]	0/592/8	54,227	0.00%
Beam width	25	872.438	0.000 [0, 0]	0/600/0	84,038	2.00%
Beam width	100	872.438	0.000 [0, 0]	0/600/0	93,174	6.17%
Beam width	200	872.438	0.000 [0, 0]	0/600/0	93,664	7.00%
Candidate cap	15/10	674.560	−197.878 [−232.487, −164.070]	3/104/493	5,547	0.00%
Candidate cap	60/40	984.867	112.428 [62.183, 171.164]	362/151/87	362,108	48.50%
Candidate cap	uncapped	706.685	−165.753 [−218.332, −107.706]	181/122/297	542,458	67.67%
Reward coefficient	5,000	872.388	−0.050 [−0.132, 0]	0/596/4	90,879	5.50%
Reward coefficient	20,000	872.465	0.027 [0, 0.083]	4/596/0	91,010	5.00%
SPH coefficient	0 / 2.5 / 10	872.438	0.000 [0, 0]	0/600/0	91,499–91,609	5.00–5.17%
Effective-time coefficient	0 / 0.05 / 0.2	872.438	0.000 [0, 0]	0/600/0	91,520–91,608	4.83–5.17%
Negative-penalty scale	0.5	872.465	0.027 [0, 0.083]	4/596/0	91,579	4.67%
Negative-penalty scale	2	872.375	−0.063 [−0.147, 0]	0/595/5	96,847	0.00%

Panel C: Confirmatory width–cap interaction (72-task fixed subset)

Width	Cap	Mean score	Change from subset default	Mean labels	CapHit
25	15/10	607.306	−200.958	4,507	0.00%
50	15/10	607.306	−200.958	5,461	0.00%
100	15/10	607.306	−200.958	6,492	0.00%
25	30/20	808.264	0.000	56,077	0.00%
50	30/20	808.264	0.000	61,267	0.00%
100	30/20	808.264	0.000	62,404	0.00%
25	60/40	1014.597	206.333	464,867	27.78%
50	60/40	982.306	174.042	328,797	40.28%
100	60/40	1011.278	203.014	511,888	38.89%

Source: `paper/results/exp3_parameter_sensitivity/sensitivity_canonical_summary_20260727.csv`

Source: `paper/results/exp3_parameter_sensitivity/sensitivity_canonical_paired_vs_default_20260727.csv`

Source: `paper/results/exp3_parameter_sensitivity/sensitivity_canonical_interaction_20260727.csv`

Supplementary Table S6. Route-case numerical details

Case	Algorithm	Stop Score	Generation (ms)	Score/hour	Effective min	Stops
A	ReRAPTOR	1455	13,831.11	186.1	469	21
A	NSGA-II	684	31,749.84	87.1	471	11
A	VNS	508	103.20	64.3	474	10
B	ReRAPTOR	375	115.79	192.3	117	6
B	Pure RAPTOR	376	10,918.73	201.4	112	6
B	Greedy	263	0.85	131.5	120	4
C	ReRAPTOR	1479	4,112.97	185.3	479	21
C	NSGA-II	1680	50,922.89	210.9	478	20
C	VNS	1482	829.10	185.2	480	19

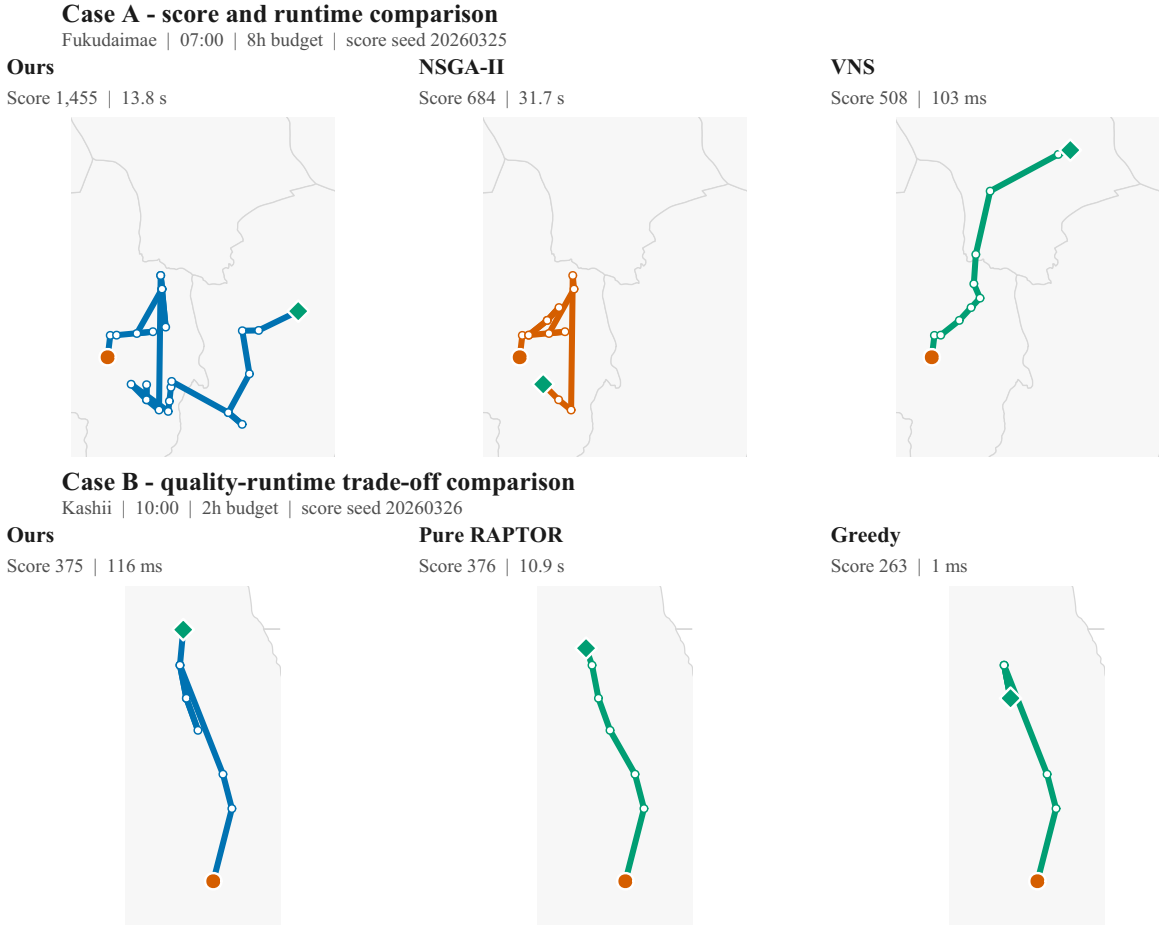
The prose comparisons are directly calculable from these rows. For example, Case B retains $375/376 \times 100 = 99.73\%$ of Pure RAPTOR's score and is $10918.73/115.79 = 94.30$ times faster. The reward per stop in Case C is $1680/20 = 84.00$ for NSGA-II and $1479/21 = 70.43$ for ReRAPTOR.

Source: `paper/figures/exp4/exp4_case_summary.md`

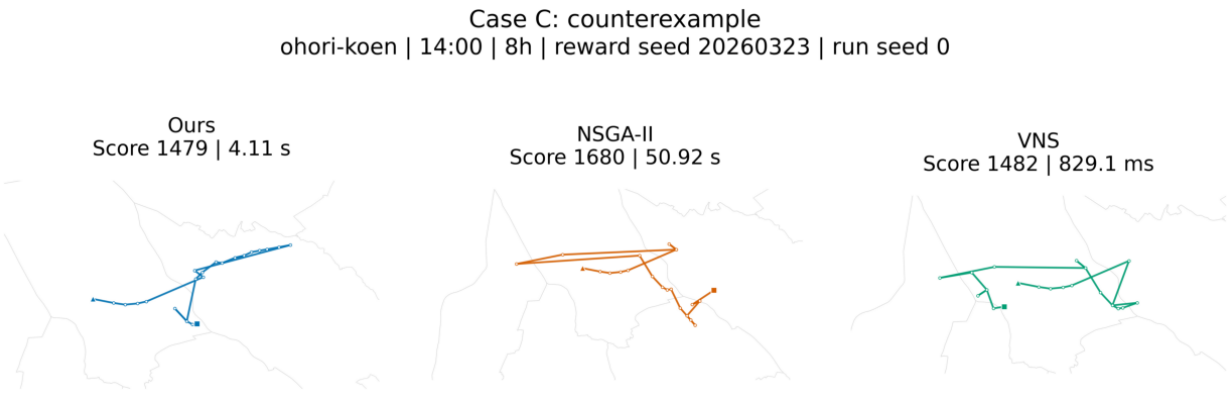
Source: `paper/results/exp4_route_visualization/exp4_cases.json`

Supplementary Figure S1. Fixed route-case selection and route maps

Case A is the largest query-level lead for which ReRAPTOR is both higher-scoring and faster than NSGA-II. Case B is the smallest score deficit among queries where ReRAPTOR ranks first by ideal-point distance. Case C is the largest ReRAPTOR deficit among faster positive-score queries with a ReRAPTOR/NSGA-II score ratio of at least 0.8. The panels below show the replayed routes selected under these rules; they are explanatory cases rather than new statistical observations.



Supplementary Figure S1 (continued). Counterexample map



Source: `paper/figures/exp4/exp4_main_cases.pdf`

Source: `paper/figures/exp4/exp4_appendix_cases.pdf`

Source: `paper/results/exp4_route_visualization/exp4_cases.json`

Supplementary Table S7. Fixed baseline configurations

Method	Fixed configuration used in Experiment 1	Executable specification
NSGA-II	Population 100; at most 200 generations or 60 s; binary tournament, nondominated sorting and crowding; timetable-feasible crossover; tail-regrowth mutation probability 0.22; seeds 0–4.	`paper/experiments/algorithms/nsga2.py`; constants in `paper/experiments/common/config.py`.
ILS+2-opt	Timetable-feasible cut-and-tail regrowth; 8–26 local trials per round; stop after 28 non-improving rounds or 60 s; seeds 0–4.	`paper/experiments/algorithms/ils_2opt.py`.
VNS	Four neighborhood strengths; at most four repeated tail perturbations; stop after 32 stagnant rounds or 60 s; seeds 0–4.	`paper/experiments/algorithms/vns.py`.
SA	Initial temperature 1000; cooling 0.995; tail-regrowth neighborhood; stop below 10^{-3}, after 600 stagnant iterations, or after 60 s; seeds 0–4.	`paper/experiments/algorithms/sa.py`.
McRAPTOR	Sample 50 destinations per seed, run the simple multicriteria RAPTOR journey query, retain the resource-feasible route with highest benchmark search score; seeds 0–4.	`paper/experiments/algorithms/mcraptor.py`.
Pure RAPTOR	ReRAPTOR scheduled-edge expansion with candidate trimming, bucket beam, Dominance, and Negative down-ranking disabled; 60-s guard; deterministic.	`paper/experiments/algorithms/pure_raptor.py`.
Greedy ($L = 3$)	Score up to nine immediate candidates; add the best three one-step onward candidates with discount 0.3; 60-min immediate and 30-min onward lookahead; deterministic.	`paper/experiments/algorithms/greedy.py`.
ALNS (adapted)	Four destroy–repair operators; 160 iterations or 45 s; operator-weight update every 12 iterations; $T_0 = 6$, cooling 0.985, floor 0.6; seeds 0–4.	`paper/experiments/algorithms/zhou_wang_2025_alns_adapted.py`.
DynPref (adapted)	Utilization thresholds 0.35 and 0.70; exploration rates 0.50/0.38/0.22; stop after 24 failed repairs or 45 s; seeds 0–4.	`paper/experiments/algorithms/zhang2025_dynamic_pref_adapted.py`.

All methods use the same timetable-feasibility checker, three-minute transfer buffer, Stop Score map, per-stop effective-time charge, and post-return validator. Deterministic methods run once per matched task; stochastic methods are averaged over seeds 0–4.

Source: `paper/results/exp1_algorithm_comparison/exp1_results_canonical_final_10alg_manuscript_20260723.json`

Source: `paper/experiments/common/config.py`